



Q.1. Give short answers of the following:

(6x5=30)

1. Convert the Laguerre equation $xy'' + (1-x)y' + \alpha y = 0$ into a self adjoint form.

2. Prove that convolution of two functions is commutative.

3. By using the Generating function for Hermite polynomial, derive the recurrence relation

$$H_{n+1}(x) = 2xH_n(x) - 2nH_{n-1}(x)$$

4. Show that $\Gamma\left(\frac{1}{2} + n\right) = \frac{(2n-1)!!}{2^n} \sqrt{\pi}$

5. Show that $J_{3/2}(x) = J_{\frac{3}{2}}(x) = \sqrt{\frac{2}{\pi x}} \left(\frac{\sin x - x \cos x}{x} \right)$

6. A function $f(x)$ is expanded in an exponential Fourier series. Show that

$$f(x) = \sum_{-\infty}^{\infty} c_n e^{inx}$$

Q.2 Answers the following questions.

(5x6=30)

1. Find the Fourier series expansion of the function $f(x) = \begin{cases} x, & 0 \leq x < \pi \\ x - 2\pi, & \pi < x \leq 2\pi \end{cases}$

2. Find the Green's function for is $\mathcal{L}y = -\frac{d^2y}{dx^2} = f(x)$ subject to the boundary condition $y(0) = y(1) = 0$ and solve this O.D.E for $f(x) = \sin \pi x$

3. Verify the Dirac delta function

$$\delta(1-x) = \sum_{l=0}^{\infty} \frac{2l+1}{2} P_l(x)$$

4. Show that recurrence relation $\frac{d}{dx} [J_n(x)] = \frac{1}{2} [J_{n-1}(x) - J_{n+1}(x)]$ flows directly from differentiation of $J_n(x) = \frac{1}{\pi} \int_0^\pi \cos(n\theta - x \sin \theta) d\theta$.

5. Find the Fourier transform of $\left(\frac{1}{r^2}\right)$.

MMP-I

Date: _____

Past Paper 2023 (solution)

Qno 1

Short Answers

1. Convert Legendre eq.

 $xy'' + (1-x)y' + ay = 0$ into self adjoint

$$xy'' + (1-x)y' + ay = 0$$

'X' b.s by e^{-x}

$$xe^{-x} \frac{d^2y}{dx^2} + (1-x)e^{-x} \frac{dy}{dx} + e^{-x} ay = 0$$

$$P_0 = xe^{-x}, \quad P_1 = (1-x)e^{-x}$$

Diff. w.r.t x

$$P_0' = e^{-x} + (-1)xe^{-x} = e^{-x}(1-x)$$

self adjoint by multiplying e^{-x}

$$\text{So, } w(x) = e^{-x}$$

$$\int_{-\infty}^{\infty} e^{-x} L_m(x) L_n(x) dx = \delta_{mn}$$

PUACP

2. Prove convolution of two function is commutative

if f, g are defined for $t \geq 0$
then convolution of f and g is

$$(f * g)(t) = \int_0^t f(\tau) g(t-\tau) d\tau$$

for commutative property

$$f * g = g * f$$

So let $u = t - \tau \Rightarrow \tau = t - u$

$$du = -d\tau$$

$$-du = d\tau$$

$$\text{as } \tau = t \Rightarrow u = 0$$

$$\tau = 0 \Rightarrow u = t$$

Putting in Q

$$(f * g) = - \int_t^0 f(t-u) g(u) du$$

$$(f * g) = \int_0^t g(u) f(t-u) du$$

as u is dummy variable

$$(f * g) = \int_0^t g(\tau) f(t-\tau) d\tau = (g * f)$$

3.

$$H_{n+1}(x) = 2x H_n(x) - 2n H_{n-1}(x)$$

for Hermite polynomials

$$\sum_{n=0}^{\infty} \frac{H_n(x) t^n}{n!} = e^{2tx - t^2}$$

Diff. w.r.t 't'

$$\sum_{n=0}^{\infty} \frac{H_n(x) n t^{n-1}}{n!} = e^{2tx - t^2} (2x - 2t)$$

$$\sum_{n=0}^{\infty} \frac{H_n(x) n t^{n-1}}{n!} = 2x \sum_{n=0}^{\infty} \frac{H_n(x) t^n}{n!} - 2 \sum_{n=0}^{\infty} \frac{H_n(x) t^{n+1}}{(n+1)!}$$

if $n \rightarrow n+1$ in L.H.S and $n \rightarrow n-1$ in 2nd term of R.H.S

$$\sum_{n=0}^{\infty} \frac{H_{n+1}(x) (n+1) t^n}{(n+1)!} = 2x \sum_{n=0}^{\infty} \frac{H_n(x) t^n}{n!} - 2 \sum_{n=0}^{\infty} \frac{H_n(x) t^n}{(n-1)!}$$

comparing coeff. of t^n

$$\frac{H_{n+1}(x) (n+1)}{(n+1)!} = \frac{2x H_n(x)}{n!} - \frac{2 H_{n-1}(x)}{(n-1)!}$$

$$\frac{H_{n+1}(x)}{(n+1)n!} = \frac{2x H_n(x)}{n!} - \frac{2 H_{n-1}(x)}{n(n-1)!}$$

$$\frac{H_{n+1}(x)}{n!} = \frac{2x H_n(x)}{n!} - \frac{2n H_{n-1}(x)}{n!}$$

$$H_{n+1}(x) = 2x H_n(x) - 2n H_{n-1}(x)$$

4. Show that $\Gamma\left(\frac{1}{2} + n\right) = \frac{(2n-1)!!}{2^n} \sqrt{\pi}$

$$\because \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

$$\text{if } (2n)!! = 2^n n!$$

$$= 2^n \Gamma(n+1)$$

$$(2n-1)!! = 2^n \Gamma\left(n + \frac{1}{2}\right)$$

$$\Gamma\left(n + \frac{1}{2}\right) = \frac{(2n-1)!!}{2^n}$$

$$= \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2^n} \Gamma\left(\frac{1}{2}\right)$$

$$\Gamma\left(n + \frac{1}{2}\right) = \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2^n} \sqrt{\pi}$$

For all product of even and odd functions

$2n$ or $(2n-1)$ we use double factorial

$$\text{So, } \Gamma\left(n + \frac{1}{2}\right) = \frac{(2n-1)!!}{2^n} \sqrt{\pi}$$

5. Show that

$$J_{3/2}(x) = J_{\frac{3}{2}}(x) = \sqrt{\frac{2}{x\pi}} \left(\frac{\sin x - x \cos x}{x} \right)$$

from recurrence relation

$$J_{n+1}(x) = \frac{2n}{x} J_n(x) - J_{n-1}(x)$$

put $n = \frac{1}{2}$

$$J_{3/2} = \frac{1}{x} J_{1/2}(x) - J_{-1/2}(x)$$

$$= \frac{1}{x} \sqrt{\frac{2}{x\pi}} \sin x - \sqrt{\frac{2}{x\pi}} x \cos x$$

$$= \sqrt{\frac{2}{x\pi}} \left(\frac{\sin x}{x} - x \cos x \right) = \sqrt{\frac{2}{x\pi}} \left(\frac{\sin x - x \cos x}{x} \right)$$

$$J_{3/2}(x) = \sqrt{\frac{2}{x\pi}} \left(\frac{\sin x - x \cos x}{x} \right)$$

where

$$J_{1/2}(x) \Rightarrow J_n(x) = \frac{x^n}{2^n \Gamma(n+1)} \left\{ 1 - \frac{x^2}{2(2n+2)} + \dots \right\}$$

putting $n = \frac{1}{2}$

$$J_{1/2}(x) = \frac{\sqrt{x}}{\sqrt{2} \Gamma(\frac{1}{2}+1)} \left\{ 1 - \frac{x^2}{2(1+2)} + \dots \right\}$$

$$= \frac{\sqrt{x}}{\sqrt{2} \cdot \frac{1}{2} \sqrt{\pi}} \left\{ 1 - \frac{x^2}{3 \cdot 2 \cdot 1} + \dots \right\}$$

$$= 2 \sqrt{\frac{x}{2\pi}} \left\{ 1 - \frac{x^2}{3!} + \dots \right\}$$

$$= \frac{1}{x} \sqrt{\frac{2x}{\pi}} \left\{ x - \frac{x^3}{3!} + \dots \right\}$$

$$J_{1/2}(x) = \sqrt{\frac{2}{x\pi}} \sin x$$

Similarity $J_{-1/2}(x) = \sqrt{\frac{2}{x\pi}} \cos x$

b. $f(x) = \sum_{n=-\infty}^{+\infty} c_n e^{inx}$

The function will be real if

$$c_n e^{inx} + c_{-n} e^{-inx} = 0 \text{ is real}$$

$$\text{So, } c_n e^{inx} + c_{-n} e^{-inx} = c_n (\cos nx + i \sin nx) + c_{-n} (\cos nx - i \sin nx)$$

$$= (c_n + c_{-n}) \cos nx + (c_n - c_{-n}) i \sin nx$$

$$c_n e^{inx} + c_{-n} e^{-inx} = \text{Real } \cos nx + \text{Imag } \sin nx$$

$$c_n = a_n + i b_n$$

equivalent require $c_n^* = c_{-n}$

Q2. Long Questions.

1.

Find Fourier Series expansion

A

$$J_{3/2}(x) = \sqrt{\frac{2}{\pi x}} \left(\frac{5 \sin x - x \cos x}{x} \right)$$

from recurrence relation

$$J_{n+1}(x) = \frac{2n}{x} J_n(x) - J_{n-1}(x)$$

Put $n = 1/2$

$$J_{3/2} = \frac{1}{x} J_{1/2}(x) - J_{-1/2}(x)$$

$$= \frac{1}{x} \sqrt{\frac{2}{\pi x}} \sin x - \sqrt{\frac{2}{\pi x}} \cos x$$

$$= \sqrt{\frac{2}{\pi x}} \left(\frac{\sin x}{x} - \cos x \right) = \sqrt{\frac{2}{\pi x}} \left(\frac{5 \sin x - x \cos x}{x} \right)$$

where

$$\therefore J_{1/2}(x) \Rightarrow J_n(x) = \frac{x^n}{2^n \Gamma(n+1)} \left\{ 1 - \frac{x^2}{2(2n+2)} + \frac{x^4}{2 \cdot 4(2n+2)(2n+4)} + \dots \right\}$$

Putting $n = 1/2$

$$J_{1/2}(x) = \frac{\sqrt{x}}{\sqrt{2} \Gamma(1/2+1)} \left[1 - \frac{x^2}{2(1+2)} + \dots \right]$$

$$= \frac{1}{\pi} \left[\left\{ \frac{x \sin nx}{n} + \frac{\cos nx}{n^2} \right\} \right]_0^{\pi} + \left\{ \frac{(x-2\pi) \sin nx}{n} + \frac{\cos nx}{n^2} \right\} \Big|_{\pi}^{2\pi}$$

$$= \frac{1}{\pi} \left[\left(0 + \frac{(-1)^n}{n^2} - \frac{1}{n^2} \right) + \left(0 + \frac{1}{n^2} + 0 - \frac{(-1)^n}{n^2} \right) \right]$$

$$= \frac{1}{\pi} \left(\frac{(-1)^n - 1}{n^2} + \frac{1 - (-1)^n}{n^2} \right) = \frac{1}{n^2 \pi} \left((-1)^n - 1 + 1 - (-1)^n \right)$$

$$\boxed{a_n = 0}, \quad b_n = \frac{1}{\pi} \int_a^b f(x) \sin nx \, dx$$

$$b_n = \frac{1}{\pi} \left[\int_0^{\pi} x \sin nx \, dx + \int_{\pi}^{2\pi} (x-2\pi) \sin nx \, dx \right]$$

$$= \frac{1}{\pi} \left[\left\{ \frac{x \cos nx}{n} - \frac{\cos nx}{n^2} \right\} \right]_0^{\pi} + \left\{ \frac{(x-2\pi) \cos nx}{n} - \frac{\cos nx}{n^2} \right\} \Big|_{\pi}^{2\pi}$$

$$= \frac{1}{\pi} \left[\left(\frac{\pi \cos n\pi}{n} - \frac{\cos n\pi}{n^2} \right) - \left(0 - \frac{\cos 0}{n^2} \right) + \left(0 - \frac{\cos 2n\pi}{n^2} \right) - \left(\frac{-\pi \cos n\pi}{n} - \frac{\cos n\pi}{n^2} \right) \right]$$

$$= \frac{1}{\pi} \left[\left(\frac{\pi \cos n\pi}{n} - \frac{\cos n\pi}{n^2} \right) - \left(0 - \frac{\cos 0}{n^2} \right) + \left(0 - \frac{\cos 2n\pi}{n^2} \right) - \left(\frac{-\pi \cos n\pi}{n} - \frac{\cos n\pi}{n^2} \right) \right]$$

$$= \frac{1}{\pi} \left[\frac{\pi(-1)^n}{n} - \frac{(-1)^n}{n^2} + \frac{1}{n^2} - \frac{1}{n^2} + \frac{\pi(-1)^n}{n} + \frac{(-1)^n}{n^2} \right]$$

$$\boxed{b_n = \frac{2(-1)^n}{n}}$$

MTWTF S

Date: _____

$$f(x) = \frac{3\pi}{4} + \sum_{n=1}^{\infty} \frac{2(-1)^n}{n} \sin nx$$

$$f(x) = \frac{3\pi}{4} + [-2 \sin x + \sin 2x - 3 \sin 3x + \dots]$$

$$J_{\frac{3}{2}}(x) = \sqrt{\frac{2}{\pi x}} \left(\frac{\sin x - x \cos x}{x} \right)$$

2. Find Green's function of
 $dy = -\frac{d^2 y}{dx^2} = f(x)$ boundary

condition $y(0) = y(1) = 0$ --- ?

let operator α ie

$$L G(x, x^3) = \delta(x - x^3)$$

$$\text{For } x \neq x^3 \Rightarrow \delta(x - x^3) = 0 \Rightarrow x \neq x^3$$

$$\alpha G(x, x^3) = 0$$

$$\alpha = -\frac{d^2}{dx^2} - \frac{1}{4}$$

$$\left(-\frac{d^2}{dx^2} - \frac{1}{4}\right) = 0 \Rightarrow D^2 = -\frac{1}{4}$$

$$\boxed{D = \pm \frac{i}{2}}$$

$$\text{b.c.s} \Rightarrow G(x, x^3) = A \sin\left(\frac{x}{2}\right) + B \cos\left(\frac{x}{2}\right)$$

Case I $x < x^3$

$$G_1(x, x^3) = A_1 \sin \frac{x}{2} + B_1 \cos \frac{x}{2}$$

$$\left. \begin{array}{l} \text{Boundary} \\ \text{condition} \end{array} \right\} \begin{array}{l} A_1 \sin \frac{0}{2} + B_1 \cos \frac{0}{2} = 0 \\ \boxed{B_1 = 0} \end{array}$$

$$G_1(x, x^3) = A_1 \sin \frac{x}{2}$$

$$= C \sin \frac{x}{2}, \quad A_1 = C$$

Case II $x > x^3$

$$G_6(x, x^3) = A_2 \sin \frac{x}{2} + B_2 \cos \frac{x}{2}$$

Applying Boundary conditions

$$A_2 \sin(\pi/2) + B_2 \cos \pi/2 = 0$$

$$\boxed{A_2 = 0} \quad \boxed{B_2 = D}$$

$$G_6(x, x^3) = D \cos \frac{x}{2}$$

4. Show that recurrence relation

$$\frac{d}{dx} [J_n(x)] = \frac{1}{2} [J_{n-1}(x) - J_{n+1}(x)]$$

follows directly from

$$J_n(x) = \frac{1}{\pi} \int_0^\pi \cos(n\theta - x \sin \theta) d\theta$$

$$J_n(x) = \frac{1}{\pi} \int_0^\pi \cos(n\theta - x \sin \theta) d\theta$$

differentiate w.r.t 'x'

$$J'_n(x) = \frac{1}{\pi} \int_0^\pi -\sin(n\theta - x \sin \theta) (\sin \theta) d\theta$$

$$= \frac{1}{2\pi} \int_0^\pi -2 \sin(n\theta - x \sin \theta) \sin \theta d\theta$$

By formula

$$-2 \sin \alpha \sin \beta = \cos(\alpha + \beta) - \cos(\alpha - \beta)$$

MUTWTF S

$$2 \sin \alpha \sin \beta = \cos(\alpha - \beta) - \cos(\alpha + \beta)$$

So,

$$J_n(x) = \frac{1}{2\pi} \int_0^\pi \cos(n\theta - x \sin \theta) d\theta - \cos(n\theta + x \sin \theta) d\theta$$

$$= \frac{1}{2\pi} \int_0^\pi \cos[(n-1)\theta - x \sin \theta] - \cos[(n+1)\theta - x \sin \theta] d\theta$$

(2)

As;

$$J_{n-1}(x) = \frac{1}{\pi} \int_0^\pi \cos[(n-1)\theta - x \sin \theta] d\theta$$

$$J_{n+1}(x) = \frac{1}{\pi} \int_0^\pi \cos[(n+1)\theta - x \sin \theta] d\theta$$

$$\text{So, } J_n(x) = \frac{1}{2} [J_{n-1}(x) - J_{n+1}(x)]$$

مكتبة جامعة

PUACP

5.

$$F(x) = \frac{1}{x^2}$$

Fourier transform of function

$$F(s) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-isx} f(x) dx$$

In 3D

$$F(\vec{s}) = \frac{1}{\sqrt{(2\pi)^3}} \iiint_{\vec{s}} f(\vec{x}) e^{i\vec{s}\vec{x}}$$

using Polar coordinates

$$= \frac{1}{(2\pi)^{3/2}} \int_0^\alpha \int_0^\pi \int_0^{2\pi} \frac{1}{r^4} e^{i\vec{s}\cdot\vec{r}} r^2 \sin\theta \, d\varphi \, d\theta \, dr$$

$$= \frac{1}{(2\pi)^{3/2}} \int_0^\alpha \int_0^\pi e^{i s r \cos\theta} \sin\theta \int_0^{2\pi} (1) \, d\varphi \, d\theta \, dr$$

$$= \frac{1}{(2\pi)^{3/2}} \int_0^\alpha \int_0^\pi -2\pi e^{i s r \cos\theta} \sin\theta \, d\theta \, dr$$

$$= \frac{-2\pi}{(2\pi)^{3/2}} \int_0^\alpha \left[\int_0^\pi e^{i s r \cos\theta} \sin\theta \, d\theta \right] dr$$

$$= \frac{-1}{\sqrt{2\pi}} \int_0^\alpha \left[\frac{e^{i s r \cos\theta}}{i s r} \right] dr = \frac{-1}{\sqrt{2\pi}} \int_0^\alpha \frac{(e^{i s r} - e^{-i s r})}{i s r} dr$$

if $\sin s r = \frac{e^{i s r} - e^{-i s r}}{2i}$

$$= \frac{1}{\sqrt{2\pi} i s} \int_0^\alpha \frac{2 i \sin s r}{r} dr$$

$$= \frac{2}{\sqrt{2\pi} s} \left[\int_0^\alpha \frac{\sin s r}{r} dr \right]$$

let $s r = z \Rightarrow dr = \frac{dz}{s}$

$$= \frac{2}{s\sqrt{2\pi}} \int_0^\alpha \frac{\sin z}{z} dz = \frac{2}{s\sqrt{2\pi}} \cdot \frac{\pi}{2}$$

MTWTFSS

Date: _____

$$F(s) = \sqrt{\frac{\pi}{2s}}$$

